

- 4 (a) The direction of the object changes continuously and hence there is a change in velocity of the object. By Newton's second law, there must be a force acting on the object. M1

Since the speed remains unchanged, this force cannot have a component tangential to the circle/ does no work. This implies that the force is perpendicular to the velocity/displacement of object acting towards centre of circle. C1

- (b) i For the block to just make it through the top of the loop, the weight is just providing for the centripetal force.

$$mg = mv_{\text{top}}^2 / R \quad \text{C1}$$

$$(0.50)(9.81) = (0.50)(v_{\text{top}}^2) / 1.5$$

$$v_{\text{top}} = 3.84 \text{ m s}^{-2} \quad [1, \text{ also award if student leave as } KE = 3.68 \text{ J}] \quad \text{A1}$$

- ii By Conservation of energy, C1

$$KE_i + EPE_i + GPE_i - \text{work done by friction} = KE_f + EPE_f + GPE_f$$

$$0 + \frac{1}{2} ke^2 + 0 - Fd = \frac{1}{2} mv_{\text{top}}^2 + 0 + mg(2R)$$

$$\frac{1}{2} (78.4)(e^2) - (1.47)(2.5) = \frac{1}{2} (0.50)(3.836^2) + 0.50(9.81)(2 \times 1.5) \quad \text{A1}$$

Solving, $e = 0.750 \text{ m}$

- (c) v_{min} remains unchanged. A1

At C, to stay in contact, centripetal acceleration of the block = acceleration of C1

free fall, $\frac{v_{\text{min}}^2}{r} = g$ Hence $v_{\text{min}} = \sqrt{rg}$, independent of the mass of the block.

5. 2. free: oscillation without any loss of energy / no external forces e.g. resistive forces $\Rightarrow$